

Partial Derivative Notation:

If $z = f(x, y)$; then the partial derivatives f_x and f_y are also denoted by the symbols:

$$\frac{\partial f}{\partial x}, \frac{\partial z}{\partial x} \quad \text{and} \quad \frac{\partial f}{\partial y}, \frac{\partial z}{\partial y}.$$

Some typical notations for the partial derivatives of $z = f(x, y)$ at a point (x_0, y_0) are:

$$\left. \frac{\partial f}{\partial x} \right|_{x=x_0, y=y_0}; \quad \left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)}; \quad \left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}.$$

$$\frac{\partial f}{\partial x}(x_0, y_0); \quad \frac{\partial z}{\partial x}(x_0, y_0).$$

Example: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ for:

(i) $z = x^4 \sin(xy^3)$

(ii) $z = x^2 + xy + y^3$

Solution: Given that, $z = x^4 \sin(xy^3)$

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{\partial}{\partial x} (x^4 \sin(xy^3)) \\ &= 4x^3 \cdot \sin(xy^3) + x^4 \cdot \cos(xy^3) \cdot y^3 \\ &= 4x^3 \sin(xy^3) + x^4 y^3 \cos(xy^3). \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial y} &= \frac{\partial}{\partial y} (x^4 \sin(xy^3)) \\ &= x^4 \cdot \cos(xy^3) \cdot 3xy^2 \\ &= 3x^5 y^2 \cos(xy^3). \end{aligned}$$

⑪ Given that: $Z = x^2 + xy + y^3$

$$\frac{\partial Z}{\partial x} = 2x + y$$

$$\frac{\partial Z}{\partial y} = x + 3y^2$$

Problem: Evaluate f_x and f_y for:

(a) $f(x, y) = x^2 + xy + y^4$

(b) $f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$

(c) $f(x, y) = 3x^3y^2 + 2y^2 + 4x$

Higher Order Partial Derivatives:

Differentiate twice with respect to x :

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = f_{xx}$$

Differentiate twice with respect to y :

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = f_{yy}$$

Differentiate first with respect to x and then with respect to y :

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (f_x) = f_{xy}$$

Differentiate first with respect to y and with respect to x :

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (f_y) = f_{yx}$$

Problem: Evaluate f_{xx} , f_{yy} , f_{xy} and f_{yx} for the previous problem.

Example: If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$; then evaluate f_x and f_y .

Given that: $f(x, y) = \sin\left(\frac{x}{1+y}\right)$.

$$f_x = \frac{\partial f}{\partial x} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{1}{1+y}$$

$$= \frac{1}{1+y} \cdot \cos\left(\frac{x}{1+y}\right)$$

$$f_y = \frac{\partial f}{\partial y} = \cos\left(\frac{x}{1+y}\right) \left(\frac{-x}{(1+y)^2}\right)$$

$$= -\frac{x}{(1+y)^2} \cdot \cos\left(\frac{x}{1+y}\right)$$

Example: If $f(x, y, z) = \sin(3x + yz)$; Evaluate f_{xxyz} .

Solution: Given that: $f(x, y, z) = \sin(3x + yz)$.

$$f_x = 3 \cos(3x + yz)$$

$$f_{xx} = -9 \sin(3x + yz)$$

$$f_{xxy} = -9 \cos(3x + yz) \cdot z$$

$$= -9z \cos(3x + yz)$$

$$f_{xxyz} = 9z \sin(3x + yz) \cdot y - 9 \cos(3x + yz)$$

$$= 9zy \sin(3x + yz) - 9 \cos(3x + yz)$$

Example: Show that $u(x, y) = e^x \sin y$ is a solution of Laplace's equation $u_{xx} + u_{yy} = 0$.

Solution: Given, $u(x, y) = e^x \sin y$

$$u_x = e^x \sin y$$

$$u_{xx} = e^x \sin y$$

$$u_y = e^x \cos y$$

$$u_{yy} = -e^x \sin y$$

$$\text{L.H.S.} = u_{xx} + u_{yy} = e^x \sin y - e^x \sin y = 0 = \text{R.H.S.}$$

$$\therefore \text{L.H.S.} = \text{R.H.S.} \text{ (Showed)}$$

* A function f of two variables x and y , is a rule that assigns a unique real number $f(x, y)$ to each point (x, y) in some set D in the xy -plane.

* A function f of three variables, x, y , and z is a rule that assigns a unique real number $f(x, y, z)$ to each point (x, y, z) in some set D in the three dimensional space.

Example: Let, $f(x, y) = \sqrt{y+1} + \ln(x^2 - y)$. Find $f(e, 0)$; $f(x, 0)$ and $f(e, y)$.

Solution: Given that:

$$f(x, y) = \sqrt{y+1} + \ln(x^2 - y)$$

$$f(e, 0) = \sqrt{0+1} + \ln(e^2 - 0)$$

$$= 1 + \ln e^2$$

$$= 1 + 2 \ln e$$

$$= 1 + 2 \cdot 1$$

$$= 3.$$

$$f(x, 0) = \sqrt{0+1} + \ln(x^2 - 0) = 1 + 2 \ln(x)$$

$$f(e, y) = \sqrt{y+1} + \ln(e^2 - y)$$

Problem: Find the domain of the function:

$$f(x, y) = x e^{-\sqrt{y+2}}$$

Solution: Given that: $f(x, y) = x e^{-\sqrt{y+2}}$

There is no value for which the function $f(x)$ is undefined, so; $x \in (-\infty, \infty)$. For y ,

$$\text{we set: } y+2 \geq 0 \Rightarrow y \geq -2.$$

$$\therefore y \in [-2, +\infty)$$

So, the domain of the function is all points above or on the line $y = -2$.

Example: If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$ then evaluate f_x & f_y .

Solution: Given that: $f(x, y) = \sin\left(\frac{x}{1+y}\right)$

$$f_x = \frac{\partial f}{\partial x} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{1}{1+y} = \frac{1}{1+y} \cdot \cos\left(\frac{x}{1+y}\right)$$

$$f_y = \frac{\partial f}{\partial y} = \cos\left(\frac{x}{1+y}\right) \left(\frac{-x}{(1+y)^2}\right) = -\frac{x}{(1+y)^2} \cos\left(\frac{x}{1+y}\right)$$

Example: If $f(x, y, z) = \sin(3x + yz)$, then evaluate: $f_{xx}yz$.

Solution: Given that:

$$f(x, y, z) = \sin(3x + yz)$$

$$f_x = 3 \cos(3x + yz)$$

$$f_{xx} = -9 \sin(3x + yz)$$

$$f_{xx}y = -9 \cos(3x + yz) \cdot z = -9z \cos(3x + yz)$$

$$f_{xx}xz = 9z \sin(3x + yz) \cdot y = 9yz \sin(3x + yz)$$

$$f_{xxyz} = 9zy \sin(3x+yz) - 9 \cos(3x+yz).$$

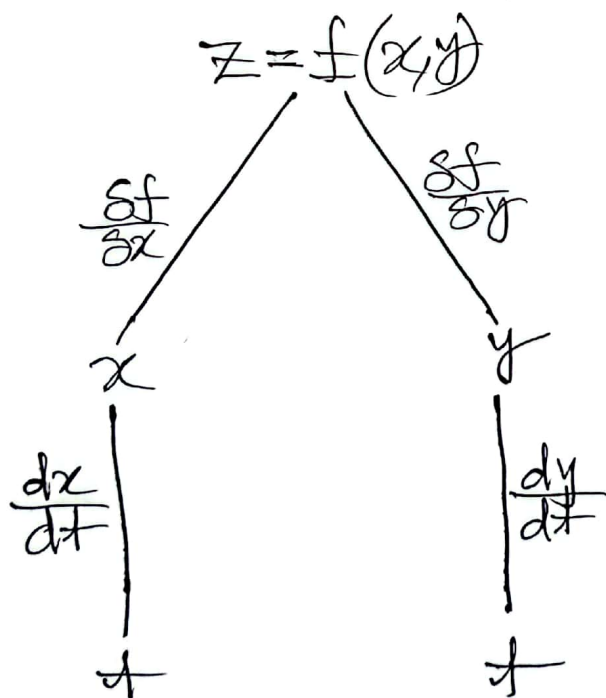
Example: Show that $u(x,y) = e^x \sin y$ is a

Q: Show that $u(x,t) = \sin(x-at)$ is a solution of wave equation: $u_{tt} = a^2 u_{xx}$.

Chain Rule for Partial Derivatives:

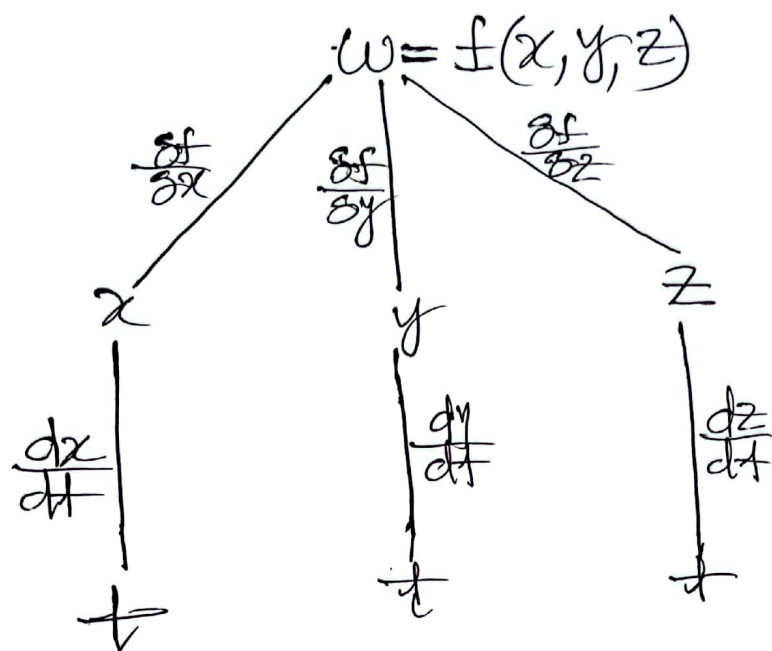
If $x = x(t)$ and $y = y(t)$ are differentiable at t and if $z = f(x,y)$ is differentiable at the point $(x,y) = (x(t), y(t))$; then $z = f(x(t), y(t))$ is differentiable at t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$



In case of three independent variable:

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$$



Problem: If $z = 3x^2y^3$; $x = t^4$ and $y = t^2$ then evaluate $\frac{dz}{dt}$.

Solution: We know that:

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \\ &= 6xy^3 \cdot 4t^3 + 9x^2y^2 \cdot 2t \\ &= 6t^4 \cdot (t^2)^3 \cdot 4t^3 + 9(t^4)^2 \cdot (t^2)^2 \cdot 2t \\ &= 24t^{13} + 18t^{13} = 42t^{13} \end{aligned}$$

Problem: If $z = x^2y$, $x = t^2$, $y = t^3$, Evaluate $\frac{dz}{dt}$.

$$\begin{aligned} \text{Solution: } \frac{dz}{dt} &= \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} \\ \therefore \frac{dz}{dt} &= 2xy \cdot 2t + x^2 \cdot 3t^2 = 2 \cdot t^2 \cdot t^3 \cdot 2t + t^4 \cdot 3t^2 \\ &= 4t^6 + 3t^6 = 7t^6 \end{aligned}$$

Problem: ① If $z = 3\cos x - \sin xy$ and $x = \frac{1}{t}$ and $y = 3t$, then evaluate $\frac{dz}{dt}$.

② If $z = xye^y$; $x = t^2$ and $y = 5t$, then evaluate $\frac{dz}{dt}$.

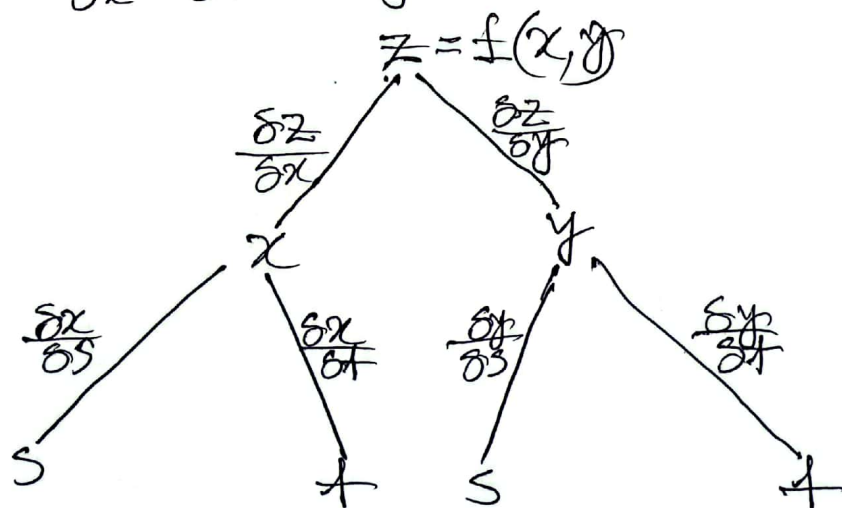
③ If $z = x^2y + xy^2$ and $x = 3t$, $y = t^2$, then evaluate: $\frac{dz}{dt}$.

Practice Problem: 13.5 $\rightarrow$ 1 to 10

The Chain Rule: Let's suppose that $z = f(x, y)$ is a differentiable function of x and y where $x = x(s, t)$ and $y = y(s, t)$ are differentiable functions of s and t . Then -

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$



Example: If $z = e^x \sin y$ where $x = s + t^2$ and $y = s^2 + t$. Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

Solution: Given: $z = e^x \sin y$.

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s} \\ &= 1^2 e^x \sin y + 2s + e^x \cos y.\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t} \\ &= 2st e^x \sin y + s^2 e^x \cos y.\end{aligned}$$

By substituting $x = s + t^2$ and $y = s^2 + t$

$$\frac{\partial z}{\partial s} = 1^2 e^{s+t^2} \sin(s^2+t) + 2s + e^{s+t^2} \cos(s^2+t).$$

$$\frac{\partial z}{\partial t} = 2st e^{s+t^2} \sin(s^2+t) + s^2 e^{s+t^2} \cos(s^2+t)$$

To find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$, we find the product of partial derivatives along each path from z to s or z to t , and then add these products.

Critical Point: A point (x_0, y_0) in the domain of a function $f(x, y)$ is called a critical point of the function if $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$ or if one or both partial derivatives do not exist at (x_0, y_0) .

(10)

Saddle point: A saddle point of a function is a point in the domain of function where it neither attains a maximum value nor attains a minimum value.

Example: Find all the Critical Points of:

$$f(x, y) = \frac{x^2}{2} - x + xy^2.$$

Solution: Given That: $f(x, y) = \frac{x^2}{2} - x + xy^2$.

$$\begin{array}{l|l} \text{Setting: } f_x(x, y) = 0 & f_y(x, y) = 0 \\ \Rightarrow x - 1 + y^2 = 0 & \Rightarrow 2xy = 0 \\ & \text{Either } x = 0 \text{ or } y = 0. \end{array}$$

$$\begin{array}{l} \text{When } x = 0; -1 + y^2 = 0 \\ \Rightarrow y = \pm 1 \end{array}$$

$$\begin{array}{l} \text{When } y = 0; x - 1 = 0 \\ \therefore x = 1. \end{array}$$

$\therefore$ The Critical Points are: $(0, 1), (0, -1), (1, 0)$.

Second Derivative Test: Let f be a function of two variables with continuous second order partial derivatives at critical points (x_0, y_0) and let:

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0).$$

a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a relative minimum at (x_0, y_0) .

b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a relative maximum at (x_0, y_0) .

c) If $D < 0$, then f has a saddle point at (x_0, y_0) .

d) If $D = 0$, then no conclusion can be drawn.

Example: Locate all relative extrema and saddle points of $f(x, y) = 4xy - x^4 - y^4$.

Solution: Given that: $f(x, y) = 4xy - x^4 - y^4$.

To find out the critical points we set.

$$f_x(x, y) = 0$$

$$\Rightarrow 4y - 4x^3 = 0$$

$$\therefore y = x^3$$

$$f_y(x, y) = 0$$

$$\Rightarrow 4x - 4y^3 = 0$$

$$\Rightarrow x = y^3$$

$$\Rightarrow x = (x^3)^3$$

$$\Rightarrow x - x^9 = 0$$

$$\Rightarrow x(1 - x^8) = 0$$

$$\Rightarrow x = 0 \text{ or } x^8 = 1$$

$$\therefore x = 0, 1, -1.$$

$$\text{When } x = 0 ; y = 0$$

$$\text{When } x = 1 ; y = 1$$

$$\text{When } x = -1 ; y = -1.$$

The only real critical points are:

$$(0, 0), (1, 1), (-1, -1).$$

Here, $f_{xx} = -12x^2$

$$f_{yy} = -12y^2$$

$$f_{xy} = 4.$$

$$D = f_{xx}f_{yy} - f_{xy}^2 = 144x^2y^2 - 16.$$

Critical Points	D	f_{xx}	Conclusion
(0,0)	$-16 < 0$		Saddle point
(1,1)	$128 > 0$	$-12 < 0$	Relative Maximum
(-1,-1)	$128 > 0$	$-12 < 0$	Relative Maximum

$\therefore (0,0)$ is saddle point and there exists relative maximum at points (1,1) and (-1,-1).
Now, $f(1,1)$ and $f(-1,-1)$ need to be calculated.

* Taylor Series of Two Variable Functions:

Let's recall the Taylor series expansion of a single variable function $f(x)$ at $x=a$.

$$f(x) = f(a) + \left. \frac{df}{dx} \right|_{x=a} (x-a) + \frac{1}{2} \left. \frac{d^2f}{dx^2} \right|_{x=a} (x-a)^2 + \dots$$

For a function of two variables $f(x, y)$; Taylor series expansion at (a, b) is:

$$\begin{aligned} f(x, y) = & f(a, b) + f_x(a, b)(x-a) + f_y(a, b)(y-b) \\ & + \frac{1}{2} (f_{xx}(a, b)(x-a)^2 + 2f_{xy}(a, b)(x-a)(y-b) \\ & + f_{yy}(a, b)(y-b)^2) + \text{terms of degree 3 or more.} \end{aligned}$$

* When you are asked to calculate the polynomial you need not put the dot sign. Because the dot sign is used to notify the series expansion.

Example: Calculate the second degree Taylor polynomial of the function $f(x, y) = \sin(2x) + \cos(y)$ near $(0, 0)$.

Solution: Given: $f(x, y) = \sin(2x) + \cos(y)$.

$$f(0, 0) = \sin 2(0) + \cos(0) = 1$$

$$f_x(x, y) = 2\cos 2x ; f_x(0, 0) = 2$$

$$f_y(x, y) = -\sin y ; f_y(0, 0) = 0$$

$$f_{xx}(x, y) = -4\sin 2x ; f_{xx}(0, 0) = 0$$

$$f_{xy}(x, y) = 0 = f_{yx}(x, y) ; f_{xy}(0, 0) = 0 = f_{yx}(0, 0).$$

$$f_{yy}(x, y) = -\cos y \quad ; \quad f_{yy}(0, 0) = -1.$$

$\therefore$ The second degree Taylor Polynomial of $f(x)$ is :

$$f(x) = 1 + 2(x-0) + 0(y-0) + \frac{1}{2}(0(x-0)^2 + 0(x-0)(y-0) + (-1)(y-0)^2).$$

$$\therefore \sin 2x + \cos y = 1 + 2x - \frac{y^2}{2}.$$

Differentiation of Vectors:

In three dimensional cartesian coordinate system, a vector is represented mathematically as, $\vec{A} = \hat{i}A_x + \hat{j}A_y + \hat{k}A_z$.

with $\hat{i}$, $\hat{j}$ and $\hat{k}$ representing unit vectors along x , y and z axis respectively. A_x , A_y and A_z are the components of $\vec{A}$. If $\vec{A}$ is a function of some variable t ,

$$\vec{A}(t) = \hat{i}A_x(t) + \hat{j}A_y(t) + \hat{k}A_z(t).$$

Differentiation with respect to t yields.

$$\frac{d\vec{A}}{dt} = \hat{i}\frac{dA_x}{dt} + \hat{j}\frac{dA_y}{dt} + \hat{k}\frac{dA_z}{dt}.$$

Differentiation Rules:

$$\frac{d}{dt}(a\vec{A}) = a \cdot \frac{d\vec{A}}{dt} + \frac{da}{dt} \cdot \vec{A} \quad [a = \text{constant}]$$

$$\frac{d}{dt}(\vec{A} \cdot \vec{B}) = \vec{A} \cdot \frac{d\vec{B}}{dt} + \frac{d\vec{A}}{dt} \cdot \vec{B}$$

$$\frac{d}{dt}(\vec{A} \times \vec{B}) = \vec{A} \times \frac{d\vec{B}}{dt} + \frac{d\vec{A}}{dt} \times \vec{B}.$$

Gradient: If f is a function of x and y , then the gradient of f is defined by:

$$\vec{\nabla} f(x, y) = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}.$$

If f is a function of x, y and z , then the gradient of f is defined by

$$\vec{\nabla} f(x, y, z) = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}.$$

Here, the operator $\vec{\nabla}$ (Nabla) is defined by:

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}.$$

Example: If a function is defined as $f(x, y) = x^2 e^y$ then find the gradient of f at $(-2, 0)$.

Solution: Given, $f(x, y) = x^2 e^y$.

$$\begin{aligned} \text{Gradient of } f, \vec{\nabla} f &= \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \\ &= \frac{\partial}{\partial x}(x^2 e^y) \hat{i} + \frac{\partial}{\partial y}(x^2 e^y) \hat{j} \\ &= 2x e^y \hat{i} + x^2 e^y \hat{j}. \end{aligned}$$

$$\text{At } (-2, 0): \vec{\nabla} f = 2(-2)e^0 \hat{i} + (-2)^2 e^0 \hat{j} \\ = -4\hat{i} + 4\hat{j}.$$

* Divergence: A vector $\vec{V}$ is defined as:

$$\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}.$$

Then the divergence of the vector $\vec{V}$ is defined as:

$$\vec{\nabla} \cdot \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (V_x \hat{i} + V_y \hat{j} + V_z \hat{k}) \\ = \frac{\partial}{\partial x} (V_x) + \frac{\partial}{\partial y} (V_y) + \frac{\partial}{\partial z} (V_z).$$

In short, the divergence of vector $\vec{V}$ is the dot product of nabla and the vector.

Example: If a vector is defined as:

$\vec{V} = 2x^3 \hat{i} + 3yz^2 \hat{j} + 3x^2 \hat{k}$; then find the divergence of the vector at $(1, 0, 1)$.

Solution: Given that: $\vec{V} = 2x^3 \hat{i} + 3yz^2 \hat{j} + 3x^2 \hat{k}$.

Then find the divergence of the vector at $(1, 0, 1)$.

$$\vec{\nabla} \cdot \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (2x^3 \hat{i} + 3yz^2 \hat{j} + 3x^2 \hat{k}) \\ = \frac{\partial}{\partial x} (2x^3) + \frac{\partial}{\partial y} (3yz^2) + \frac{\partial}{\partial z} (3x^2) \\ = 6x^2 + 3z^2$$

$$\text{At } (1, 0, 1): \vec{\nabla} \cdot \vec{V} = 6(1)^2 + 3(1)^2 = 6 + 3 = 9.$$

Curl: If a vector $\vec{V}$ is defined as:

$$\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}.$$

Then the curl of the vector $\vec{V}$ is defined as:

$$\begin{aligned}\vec{\nabla} \times \vec{V} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) - \hat{j} \left(\frac{\partial V_z}{\partial x} - \frac{\partial V_x}{\partial z} \right) + \hat{k} \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right)\end{aligned}$$

In short, the curl of a vector $\vec{V}$ is the cross product of nabla and the vector.

Example: If a vector is defined as:

$\vec{A} = 2x^3 \hat{i} + 3yz^2 \hat{j} + 4zx \hat{k}$ then find the curl of $\vec{A}$ at $(1, 2, -1)$.

Solution: Given that:

$$\vec{A} = 2x^3 \hat{i} + 3yz^2 \hat{j} + 4zx \hat{k}.$$

$$\begin{aligned}\vec{\nabla} \times \vec{A} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x^3 & 3yz^2 & 4zx \end{vmatrix} \\ &= \hat{i} \left(\frac{\partial}{\partial y} (4zx) - \frac{\partial}{\partial z} (3yz^2) \right) - \hat{j} \left(\frac{\partial}{\partial x} (4zx) - \frac{\partial}{\partial z} (2x^3) \right) \\ &\quad + \hat{k} \left(\frac{\partial}{\partial x} (3yz^2) - \frac{\partial}{\partial y} (2x^3) \right) \\ &= -6yz \hat{i} - 4z \hat{j}.\end{aligned}$$

At $(1, 2, -1)$: $\vec{\nabla} \times \vec{A} = 12\hat{i} + 4\hat{j}$.

Field: Field refers to the physical quantity of interest and the region of space where it has a value.

Directional Derivative: If the rate of change of any physical quantity depends on the direction we are moving, it is called the directional derivative.

The directional derivative represents the instantaneous rate of change of $f(x, y)$ with respect to distance in the direction of unit vector $\vec{u}$ at the point (x_0, y_0) .

If a function $f(x, y)$ is of x and y and if $\vec{u} = u_1\hat{i} + u_2\hat{j}$ is a unit vector then the directional derivative of f in the direction of $\vec{u}$ at (x_0, y_0) is denoted by $D_{\vec{u}} f(x_0, y_0)$ and is defined by: $D_{\vec{u}} f(x_0, y_0) = \vec{\nabla} f \cdot \vec{u} |_{(x_0, y_0)}$.

Example: Find the directional derivative of $f(x, y) = xy$ at $(1, 2)$ in the direction of.

$$\vec{A} = \sqrt{3}\hat{i} + \hat{j}.$$

Solution: The unit vector is defined as:

$$\vec{u} = \frac{\vec{A}}{|\vec{A}|} = \frac{\sqrt{3}\hat{i} + \hat{j}}{\sqrt{(\sqrt{3})^2 + 1^2}} = \frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}.$$

(12)

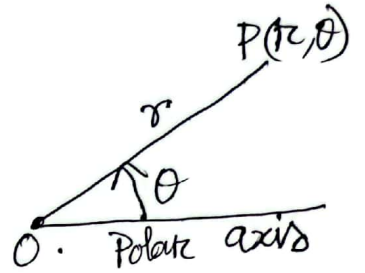
Gradient of f , $\vec{\nabla} f = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} \right) (xy)$
 $= y \hat{i} + x \hat{j}$

Directional derivative, $D_u f = (\vec{\nabla} f \cdot \vec{u})$
 $D_u f = (y \hat{i} + x \hat{j}) \cdot \left(\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j} \right)$
 $= \frac{\sqrt{3}}{2} y + \frac{1}{2} x$

At $(1, 2)$: $D_u f(1, 2) = \frac{\sqrt{3}}{2} \cdot 2 + \frac{1}{2} \cdot 1 = \sqrt{3} + \frac{1}{2}$ (Ans)

* Polar Coordinate System:

A polar coordinate system in a plane consists of a fixed point O , called the pole (or origin) and a ray emanating from the pole, called the polar axis. In such a coordinate system, we can associate with each point P in the plane a pair of polar coordinates (r, θ) where r is the distance from P to the pole and θ is an angle from the polar axis to the ray OP . The number r is called the radial coordinate of P and the number θ is the angular coordinate (or polar angle) of P .

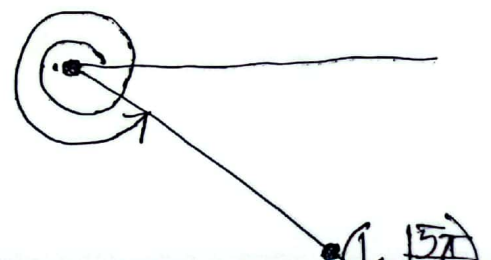
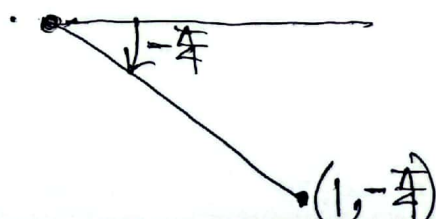
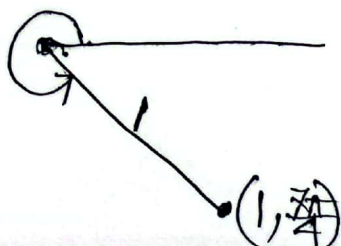


* θ is positive in the counterclockwise direction from the polar axis.

* θ is negative in the clockwise direction from the polar axis.

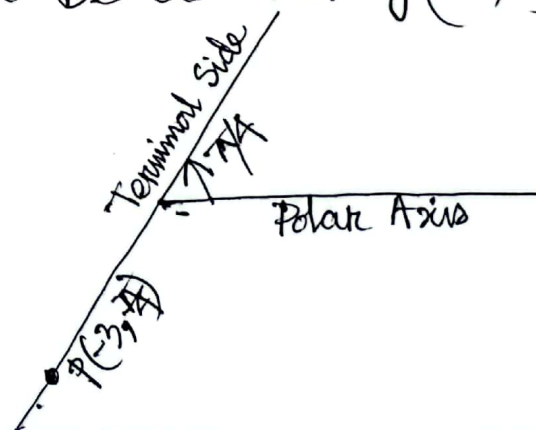
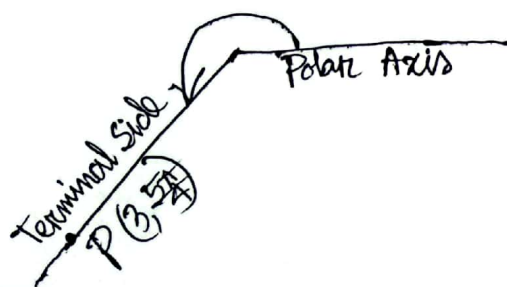
The polar coordinates of a point are not unique. For example, the polar coordinates:

$(1, \frac{7\pi}{4})$, $(1, -\frac{\pi}{4})$ and $(1, \frac{15\pi}{4})$ all represent the same point.



In general, if point P has polar coordinates (r, θ) then $(r, \theta + 2n\pi)$ and $(r, \theta - 2n\pi)$ are also polar coordinates of P for any nonnegative integer n . Thus every point has infinitely many pairs of polar coordinates.

The radial coordinate r of a point P is nonnegative, since it represents the distance from P to the pole. However, it will be convenient to allow nonnegative values for r as well. To motivate an appropriate definition, we can reach the point $P(3, \frac{5\pi}{4})$ by rotating the polar axis through an angle of $\frac{5\pi}{4}$ and then moving 3 units from the pole along the terminal side of the angle or we can reach the point P by rotating the polar axis through an angle of $\frac{\pi}{4}$ and then moving 3 units from the pole along the extension of the terminal side. This suggests that the point $(3, \frac{5\pi}{4})$ might also be denoted by $(-3, \frac{\pi}{4})$.



In general, the terminal side of the angle $\theta + \pi$ is the extension of the terminal side of θ , therefore $(-r, \theta)$ and $(r, \theta + \pi)$ are polar coordinates of the same point.

Relationship between polar and rectangular coordinate system:

Polar $(r, \theta) \rightarrow$ Rectangular (x, y)	Rectangular $(x, y) \rightarrow$ Polar (r, θ)
$x = r \cos \theta$	$r = \sqrt{x^2 + y^2}$
$y = r \sin \theta$	$\tan \theta = \frac{y}{x}$

Example: Find the rectangular coordinates of the point P whose polar coordinates are $(6, \frac{2\pi}{3})$.

Solution: We know that:

$$x = r \cos \theta = 6 \cos(\frac{2\pi}{3}) = -3$$

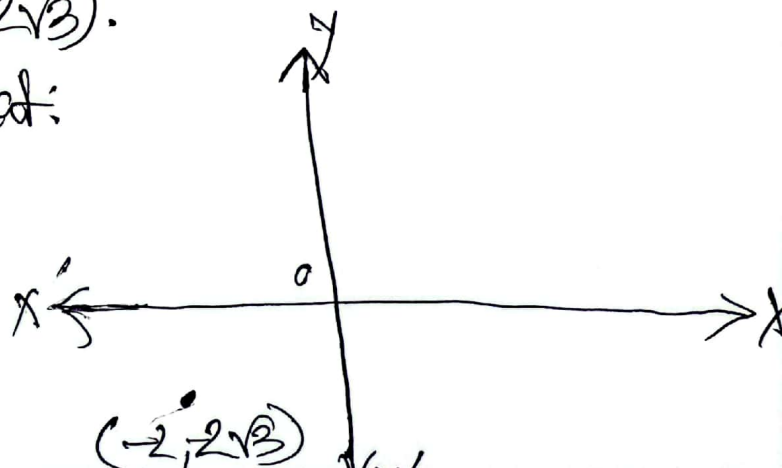
$$y = r \sin \theta = 6 \sin(\frac{2\pi}{3}) = 3\sqrt{3}$$

$\therefore$ The rectangular coordinates of P are $(-3, 3\sqrt{3})$.

Example: Find the Polar coordinates of the point P whose rectangular coordinates are $(-2, -2\sqrt{3})$.

Solution: We know that:

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{(-2)^2 + (-2\sqrt{3})^2} \\ &= \sqrt{4 + 12} = \sqrt{16} = 4 \end{aligned}$$



$$\tan \theta = \frac{y}{x}$$

$$\Rightarrow \tan \theta = \frac{-2\sqrt{3}}{-2}$$

$$\therefore \tan \theta = \sqrt{3}$$

The point $(-2, -2\sqrt{3})$ lies in the third quadrant, it follows that the angle satisfying the requirement $0 \leq \theta < 2\pi$ is $\theta = \frac{4\pi}{3}$. Thus: $(r, \theta) = (4, \frac{4\pi}{3})$.

* Transformation of Equations:

Polar $\rightarrow$ Rectangular.

$$r = \frac{6}{2 \cos \theta + 2 \sin \theta}$$

$$\Rightarrow 3r \cos \theta + 2r \sin \theta = 6$$

$$\Rightarrow 3x + 2y = 6 \quad [x = r \cos \theta \text{ and } y = r \sin \theta]$$

Rectangular $\rightarrow$ Polar.

$$9xy = 4$$

$$\Rightarrow 9r \cos \theta \cdot r \sin \theta = 4$$

$$\Rightarrow \frac{9}{2} r^2 (2 \sin \theta \cos \theta) = 4$$

$$\Rightarrow r^2 \sin 2\theta = \frac{8}{9}$$

Practice Problem: Chapter 10.2: 3, 4, 5, 9, 10, 11, 12

* Cylindrical Coordinate System:

Three coordinates are required to establish the location of a point in 3-space. In a rectangular coordinate system, the coordinates can be any real numbers, but in cylindrical and spherical coordinate systems, there are restrictions on the allowable values of the coordinates.

* Relationship between Cartesian and Cylindrical Coordinate System:

Cylindrical $\rightarrow$ Cartesian	Cartesian $\rightarrow$ Cylindrical
$(r, \theta, z) \rightarrow (x, y, z)$	$(x, y, z) \rightarrow (r, \theta, z)$
$x = r \cos \theta$	$r = \sqrt{x^2 + y^2}$
$y = r \sin \theta$	$\tan \theta = \frac{y}{x}$
$z = z$	$z = z$

Example: Find the cylindrical coordinates of point P whose ^{Cartesian} coordinates are $(4, -4, 4\sqrt{6})$.

Solution: We know that:

$$r = \sqrt{x^2 + y^2} = \sqrt{(4)^2 + (-4)^2} = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}$$

$$\tan \theta = \frac{y}{x} \Rightarrow \tan \theta = \frac{-4}{4} \Rightarrow \tan \theta = -1$$

$\Rightarrow \theta = \frac{7\pi}{4}$ [Since the given point has negative y coordinate]

$$z = 4\sqrt{6}$$

$\therefore$ The cylindrical coordinates are $(4\sqrt{2}, \frac{7\pi}{4}, 4\sqrt{6})$.

Example: Find the cylindrical coordinates of the point p whose cartesian coordinates are $(4\sqrt{2}, \frac{7\pi}{4}, 4\sqrt{6})$

Solution: We know that:

$$x = r \cos \theta = 4\sqrt{2} \cos\left(\frac{7\pi}{4}\right) = 4\sqrt{2} \cdot \frac{1}{\sqrt{2}} = 4$$

$$y = r \sin \theta = 4\sqrt{2} \sin\left(\frac{7\pi}{4}\right) = 4\sqrt{2} \left(-\frac{1}{\sqrt{2}}\right) = -4$$

$$z = z = 4\sqrt{6}$$

$\therefore$ The cylindrical coordinates are $(4, -4, 4\sqrt{6})$

Transformation of equation:

Cartesian $\rightarrow$ Cylindrical

$$2x + 3y + 4z = 1$$

$$\Rightarrow 2r \cos \theta + 3r \sin \theta + 4z = 1$$

$$\Rightarrow 4z = 1 - 2r \cos \theta - 3r \sin \theta$$

$$\therefore z = \frac{1 - 2r \cos \theta - 3r \sin \theta}{4}$$

* Spherical Coordinates:

Let's recall the cylindrical coordinates:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

Now, we substitute $r = \rho \sin \phi$ and $z = \rho \cos \phi$ in the above equation. Then we get:

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Cartesian $\rightarrow$ Spherical | Spherical $\rightarrow$ Cartesian

$$(x, y, z) \rightarrow (\rho, \phi, \theta)$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\tan \theta = \frac{y}{x}$$

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Example: Find the cartesian coordinates of the point with spherical coordinates $(4, \frac{\pi}{4}, \frac{\pi}{3})$.

Solution: We know that:

$$x = \rho \sin \phi \cos \theta$$

$$\therefore x = 4 \sin\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{3}\right) = \sqrt{2}$$

$$\therefore y = \rho \sin \phi \sin \theta = 4 \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{\pi}{3}\right) = \sqrt{6}$$

$$\therefore z = \rho \cos \phi = 4 \cos\left(\frac{\pi}{4}\right) = 2\sqrt{2}$$

The cartesian coordinates of the points are $(\sqrt{2}, \sqrt{6}, 2\sqrt{2})$.

Example: Find the spherical coordinates of the point whose cartesian coordinates are $(\sqrt{2}, \sqrt{6}, 2\sqrt{2})$.

Solution: We know that:

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{2 + 6 + 8} = 4$$

$$\cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}} = \frac{2\sqrt{2}}{4}$$

$$\Rightarrow \cos \phi = \frac{1}{\sqrt{2}} \quad \Rightarrow \phi = \frac{\pi}{4}$$

$$\tan \theta = \frac{y}{x} \quad \Rightarrow \tan \theta = \frac{\sqrt{6}}{\sqrt{2}} \quad \Rightarrow \tan \theta = \sqrt{3}$$

$$\therefore \theta = \frac{\pi}{3}$$

The spherical coordinates are $(4, \frac{\pi}{4}, \frac{\pi}{3})$.

Transformation of Equations:

Cartesian $\rightarrow$ Spherical

$$x^2 + y^2 + z^2 = 1$$

$$\Rightarrow \rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta - \rho^2 \cos^2 \phi = 1$$

$$\Rightarrow \rho^2 \sin^2 \phi (\sin^2 \theta + \cos^2 \theta) - \rho^2 \cos^2 \phi = 1$$

$$\Rightarrow \rho^2 \sin^2 \phi - \rho^2 \cos^2 \phi = 1$$

$$\Rightarrow \rho^2 (\sin^2 \phi - \cos^2 \phi) = 1$$

$$\therefore \rho^2 \cos 2\phi = -1 \quad [\cos 2\phi = -(\sin^2 \phi - \cos^2 \phi)]$$

Spherical $\rightarrow$ Cartesian

$$\rho - 2 \sin \phi \cos \theta = 0$$

$$\Rightarrow \rho^2 - 2\rho \sin \phi \cos \theta = 0$$

$$\Rightarrow \rho^2 - 2x = 0$$

$$\Rightarrow x^2 + y^2 + z^2 - 2x = 0$$

Practice Problem: Chapter 11.8:

1, 2, 3, 4, 5, 6, 7, 8, 19-26, 27-34, 35-46.

Spherical to Cylindrical:

$$r = \rho \sin \phi, \theta = \theta, z = \rho \cos \phi$$

Cylindrical to spherical:

$$\rho = \sqrt{r^2 + z^2}, \theta = \theta, \tan \phi = \frac{r}{z}$$

Example: Find the spherical coordinates of the point whose cylindrical coordinates are $(\sqrt{3}, \frac{\pi}{6}, 3)$

Solution: We know that:

$$\rho = \sqrt{r^2 + z^2} = \sqrt{(\sqrt{3})^2 + (3)^2} = \sqrt{12}$$

$$\theta = \frac{\pi}{4}$$

$$\tan \phi = \frac{r}{z} \Rightarrow \phi = \tan^{-1}\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$$

The spherical coordinates of the point is $(\sqrt{12}, \frac{\pi}{6}, \frac{\pi}{4})$.

Example: Find the cylindrical coordinates of the point whose spherical coordinates are $(5, \frac{\pi}{4}, \frac{2\pi}{3})$.